

Exp. No. 4.1

Design and Simulation of 2:4 Decoder using Verilog code in Xilinx

Aim

To design and simulate a 2-to-4 line decoder using Verilog HDL and verify its functionality using the Xilinx Vivado simulator.

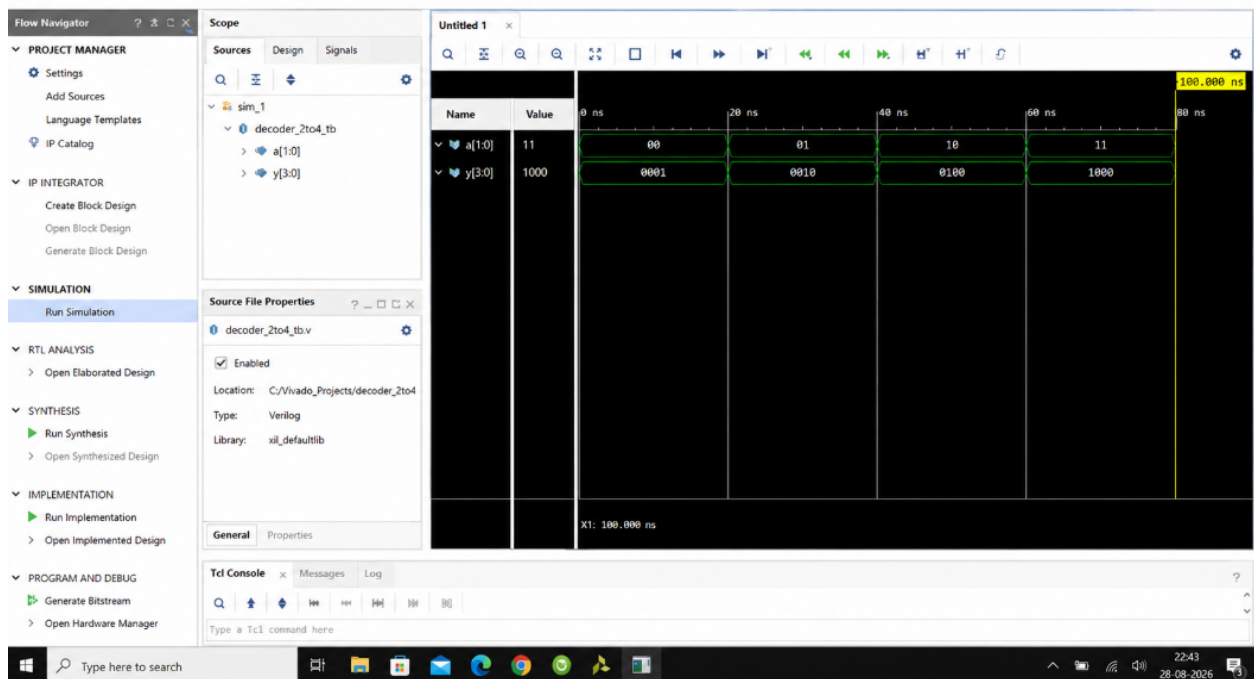
Verilog Code

```
module decoder_2to4(  
    input [1:0] a,  
    output reg [3:0] y  
);  
    always @(*)  
    begin  
        case(a)  
            2'b00: y = 4'b0001;  
            2'b01: y = 4'b0010;  
            2'b10: y = 4'b0100;  
            2'b11: y = 4'b1000;  
            default: y = 4'b0000;  
        endcase  
    end  
endmodule
```

Testbench

```
module decoder_2to4_tb;  
    reg [1:0] a;  
    wire [3:0] y;  
  
    decoder_2to4 uut (.a(a), .y(y));  
  
    initial begin  
        a = 2'b00;  
        #20 a = 2'b01;  
        #20 a = 2'b10;  
        #20 a = 2'b11;  
        #20 $finish;  
    end  
endmodule
```

Output



Result: The 2:4 decoder was designed and simulated successfully in Xilinx Vivado. For inputs a=00,01,10,11 the outputs obtained were y=0001,0010,0100,1000 respectively, matching the expected truth table.